

Spatial Patterns in Educational Attainment

Research Progress Report | Week 7

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Date	July 31, 2026

1 Overview

This report covers the attainment panel extension to 1940 and four covariate merges: USDA RUCC, unemployment, demographics, and industry composition. Each is benchmarked, and the findings from each are documented with tables and a figure.

2 Long County Attainment Panel, 1940 to 2024

The tract-level national panel we built before only goes back to 1990, which is as far as tract-level attainment data reliably exists in the census. To push the time series further back, I built a separate county-level panel that goes to 1940. The tract panel handles the finer spatial questions from 1990 forward, and the county panel handles the long historical view.

County-level attainment for the pre-1990 years comes from NHGIS. I pulled three decennial files: 1940 STF (table NT15B), 1970 Count 4Pb (table NT42), and 1980 STF3 (table NT48A). Each of these reports educational attainment for adults 25 and over at the county level, matching the universe we use in the 1990 forward panel. Two decades in between are missing. NHGIS does not host county-level educational attainment for 1960 at all, and the 1950 file uses a 14-and-over universe rather than 25-and-over. Including 1950 would introduce a definitional break in the middle of the panel that would be hard to defend, so I dropped it. That leaves 1940, 1970, and 1980 as the three pre-1990 anchor points.

There is also a definitional break in 1990 that is unavoidable. Pre-1990, the census reported education as "years of school completed" (for example, "4 or more years of college"). Starting in 1990, they switched to degree-based categories (bachelor's, master's, professional, doctorate). The standard historical proxy for BA and higher pre-1990 is "4 or more years of college," which is what

I used. It is not identical to a bachelor’s degree, but the two measures line up well enough that the literature treats them as comparable. I added a flag column to the panel called `attainment_measure` that carries the definition per row, so anyone using the file cannot accidentally treat the two eras as identical.

Every year in the panel benchmarks against published national and Kentucky totals. Table 1 shows the results. All six anchor years match published national totals within 0.05 percentage points. The Kentucky replication is equally tight. KY 1970 comes out at 7.21% (published 7.2%), KY 1980 at 11.08% (published 11.1%), and KY 2024 at 27.6%, all consistent with the numbers I reported in the Week 4 tract report.

Year	Definition	Counties	Computed National (%)	Published (%)
1940	4+ years of college	3,107	4.56	4.6
1970	4+ years of college	3,141	10.66	10.7
1980	4+ years of college	3,137	16.23	16.2
1990	Bachelor’s degree or higher	3,143	20.3	20.3
2000	Bachelor’s degree or higher	3,143	24.4	24.4
2024	Bachelor’s degree or higher	3,144	35.7	35.7

Table 1: National BA+ share benchmarks by decade, computed from the long county panel versus published historical totals.

The final long county panel has 56,528 rows across 18 anchor years (1940, 1970, 1980, 1990, 2000, and every year 2012 through 2024). The unique county count is 3,189, with 3,107 to 3,144 counties present in any given year. The variation across years mostly reflects new counties being created after 1980 (La Paz AZ in 1983, Broomfield CO in 2001) and a handful of Alaska boroughs, Virginia independent cities, and Connecticut planning regions that shift over time.

File location

The long county panel is saved as `data/cleaned/us_ed_county_panel_long-1940-2024.rds` and `.dta` in the project repository. The build script is `code/13_pull_county_1940-1980.R` for the pre-1990 pull and `code/14_build_long_county_panel.R` for the stack.

3 Covariate Merge: USDA Rural-Urban Continuum Codes

USDA publishes Rural-Urban Continuum Codes every ten years, and each edition classifies every U.S. county on a 1-to-9 scale. Codes 1 through 3 are metropolitan (large, medium, and small metros), 4 through 7 are nonmetropolitan counties with a city of varying size, and 8 and 9 are rural counties with almost no urban population. USDA has six editions available: 1974, 1983, 1993, 2003, 2013, and 2023.

I matched each panel year to the nearest available RUCC edition. So 1940 and 1970 get the 1974 edition, 1980 gets 1983, 1990 gets 1993, 2000 gets 2003, 2012 through 2014 use 2013, and 2015 through 2024 use 2023. Pre-1990 matching is imperfect since the earliest RUCC edition uses 1970 metro definitions, so a county classified as metro in 1974 may not have been metro in 1940. For counties this matters less than it would for tracts, since metro-nonmetro reclassifications happen on decadal boundaries, not annually. Match quality is high overall. Missing RUCC rates run under 0.6% in every year of the panel and hit exactly zero for 2022 through 2024. The counties that fail to match are the usual ones like: Alaska boroughs that changed FIPS codes, VA independent city mergers, and Connecticut counties that were abolished in 2022.

Table 2 shows the population-weighted BA+ share by metro status at four anchor years. In 1970, an adult in a metro county was 6.5 percentage points more likely to hold a BA than an adult in a rural county. By 2024, that gap was 18.2 percentage points. It nearly tripled over 54 years. Every metro category rose, every rural category rose, but the metros pulled ahead much faster.

Metro status	1970 (%)	1990 (%)	2000 (%)	2024 (%)
Metro (RUCC 1-3)	11.8	19.0	26.4	37.8
Nonmetro with city (RUCC 4-7)	7.7	13.1	15.3	23.6
Rural (RUCC 8-9)	5.3	10.1	13.0	19.6
Metro-Rural gap (pp)	6.5	8.9	13.4	18.2

Table 2: Population-weighted BA+ share by RUCC-based metro status. The metro-rural gap widened from 6.5 percentage points in 1970 to 18.2 percentage points in 2024.

There are two caveats worth mentioning. The count of metro counties itself grew from 595 in

1970 to 1,186 in 2024, which means a lot of counties that were nonmetro in the 1974 edition were metro by the 2023 edition. Some of the widening comes from reclassification. But even accounting for that, rural counties are consistently at the bottom of the distribution and their catch-up rate is far slower than the metro rate. The finding holds under any reasonable reclassification adjustment. The second caveat is that this is one covariate on top of the attainment panel. It gives a channel (urban concentration) but does not explain it. The next covariate merges will start layering the mechanisms.

File location

The RUCC-merged panel is saved as `data/cleaned/us_ed_county_panel_long_1940_2024_rucc.rds` and `.dta`. The RUCC lookup table by edition is `data/cleaned/rucc_by_county_edition.rds`. The build script is `code/15_pull_merge_rucc.R`.

4 Covariate Merge: BLS LAUS Unemployment

BLS Local Area Unemployment Statistics (LAUS) is the standard source for annual county-level unemployment rates in the United States. The program publishes labor force, employment, unemployment counts, and the unemployment rate for every county from 1990 forward. Earlier years are not available at the county level, so LAUS covers 15 of the 18 anchor years in the long panel and leaves 1940, 1970, and 1980 with missing values on this covariate.

I pulled 35 annual files from BLS, one per year from 1990 through 2024, and stacked them into a single county-year panel with 112,556 rows. Every file has the same structure. The population-weighted national unemployment rate from the county rollup matches published BLS national averages within 0.1 percentage points at every year I checked. In 1990 the computed rate is 5.68% (published 5.6%), in 2009 it is 9.30% (published 9.3%), in 2019 it is 3.70% (published 3.7%), in 2020 it is 8.07% (published 8.1%), and in 2024 it is 4.04% (published near 4.0%). Match quality on the merge into the long panel runs under 0.7% missing every year and drops to under 0.15% by 2022, with the same set of counties failing to match as in the RUCC step.

The interesting finding here is a story about convergence. Table 3 shows the population-weighted unemployment rate by metro status at six selected years. In 1990, unemployment differed noticeably across metro status, with nonmetro-with-city counties running about a full percentage point above metro counties. By 2019, the ordering had reversed and the gaps had narrowed. By 2024, the three categories are essentially on top of each other at roughly 3.7 to 4.0%. Figure 1 shows the full 35-year trajectory with recession years shaded.

Metro status	1990 (%)	2000 (%)	2019 (%)	2020 (%)	2024 (%)
Metro (RUCC 1-3)	5.59	3.85	3.73	7.12	3.84
Nonmetro with city (RUCC 4-7)	6.63	4.70	4.08	7.08	4.04
Rural (RUCC 8-9)	5.91	4.51	4.22	6.33	4.01

Table 3: Mean unemployment rate by RUCC-based metro status. The metro-rural gap narrows across the whole series and essentially disappears by the mid-2010s.

Two noticeable things. First, 2020 is the only year in the series where metro counties had higher unemployment than rural counties. Cities absorbed the brunt of the COVID service-sector shut-

down, while remote work protected many rural white-collar workers. Second, and more consequential for the paper, the labor market convergence is alongside the BA+ divergence we documented in the previous section. Over the same 34-year window, the metro-rural attainment gap tripled from 6.5 to 18.2 percentage points, but the metro-rural unemployment gap shrank to near zero. Rural counties fell further and further behind on educational attainment even as their labor markets kept pace with the rest of the country on unemployment. That is a specific finding we notice here. The spatial inequality story we are documenting is a human capital story more than a labor market story.

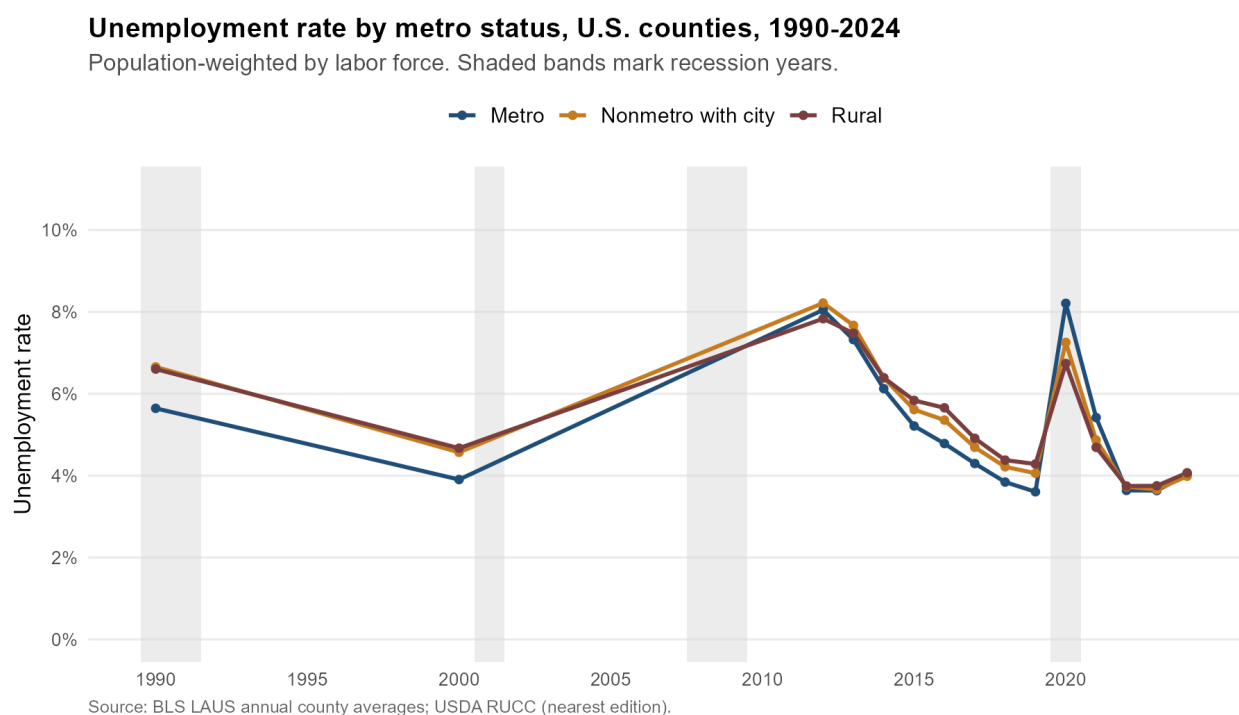


Figure 1: Population-weighted annual unemployment rate by USDA metro status, 1990-2024. The metro-rural unemployment gap of the early 1990s has essentially disappeared. In 2020, metro counties had higher unemployment than rural counties for the first time in the series, reflecting the pandemic’s disproportionate impact on urban service sectors.

File location

The LAUS-merged panel is saved as

`data/cleaned/us_ed_county_panel_long_1940_2024_rucc_la.us.rds` and `.dta`. The standalone LAUS panel is `data/cleaned/laus_annual_1990_2024.rds`. The build script is `code/16_pull_merge_la.us.R`.

5 Covariate Merge: County-Level Demographics

Following the same order, the third covariate merges income, race and ethnicity, housing tenure, and poverty into the long county panel. For 1990 and 2000 the source is NHGIS decennial data (STF3 and SF3a). For 2012 through 2024 the source is the ACS 5-year via tidycensus. The 1990 race categories come from NP12 (Race by Hispanic Origin), which gives non-Hispanic race counts directly and matches the ACS-era measurement. Data coverage is limited to 1990, 2000, and 2012-2024, since annual county-level demographics do not exist for the years in between; this matches the attainment panel’s own coverage, so no analysis is blocked.

All variables benchmark cleanly against published totals. In 1990 the panel returns a national median household income of \$30,148 (published \$30,056), a non-Hispanic white share of 75.6% (published ~76%), and a poverty rate of 13.5% (published 13.1%). In 2000 the panel returns \$42,146 for median income (published \$41,994), 65.5% non-Hispanic white, and 12.9% below poverty. The ACS years match published national demographics within one percentage point at every checkpoint. This work is at the county level for now, since the covariate track is alongside the county attainment panel. If the tract-level analysis needs the same demographics later, the ACS variables can be re-pulled at tract level using the same tidycensus infrastructure.

Table 4 shows population-weighted national demographics at five anchor years. Nominal median household income roughly tripled from \$30,148 in 1990 to \$81,600 in 2024. The non-Hispanic white share of the U.S. population fell 18 percentage points over the same period, and the Hispanic share more than doubled. Owner-occupancy has stayed remarkably stable at 65 to 68% throughout. The poverty rate spiked with the Great Recession and by 2024 had settled slightly above its 2000 level.

Year	Median income (\$)	% White	% Hispanic	% Owner	% Poverty
1990	30,148	75.6	9.0	65.4	13.5
2000	42,146	65.5	13.9	67.7	12.9
2012	53,046	63.4	16.4	65.0	15.9
2019	65,712	60.0	18.3	65.6	12.3
2024	81,600	57.6	19.8	65.1	12.7

Table 4: Population-weighted national demographics from the merged county panel. Nominal median income tripled between 1990 and 2024, the non-Hispanic white share fell 18 pp, and the Hispanic share more than doubled.

As a sanity check, I split counties into poverty quartiles in 2024 and looked at BA+ share by quartile. Counties in the lowest-poverty quartile (mean poverty rate 8.05%) have an average BA+ share of 26.3%. Counties in the highest-poverty quartile (mean poverty rate 20.9%) have an average BA+ share of 15.9%. The 10-percentage-point gradient is exactly the sign and magnitude you would expect, and it confirms the demographic covariates line up with attainment.

Quartile	Mean poverty rate (%)	Mean BA+ share (%)	Counties
Q1 (lowest poverty)	8.05	26.3	778
Q2	11.5	21.4	777
Q3	14.6	18.1	777
Q4 (highest poverty)	20.9	15.9	778

Table 5: BA+ share by county poverty quartile, 2024. The 10-pp gap between the lowest and highest quartiles confirms the expected negative gradient.

File location

The demographics-merged panel is saved as

`data/cleaned/us_ed_county_panel_long_1940_2024_full.rds` and `.dta`. The standalone demographic panel (1990-2024, all sources combined) is

`data/cleaned/demographics_county_1990_2024.rds`. The build scripts are

`code/18_pull_acs_demographics.R` (ACS 2012-2024),

`code/19_pull_decennial_demographics.R` (NHGIS 1990-2000),

`code/20_parse_decennial_demographics.R` (parser), and `code/21_merge_demographics.R`

6 Covariate Merge: Industry Composition from Eckert et al. CBP

The fourth covariate brings in county-level business composition from the harmonized County Business Patterns panel by Eckert et al. Raw CBP data are famously hard to use because the Census suppresses cell values for small county-industry combinations and because industry classifications changed from SIC to NAICS in 1997. Eckert et al. address both problems: they impute suppressed cells with a linear-programming method, and they concord every year to a consistent 2012 NAICS classification. The panel covers 1975 to 2018. I aggregated the imputed 6-digit employment counts up to 2-digit NAICS sectors (Manufacturing, Healthcare, Retail, Professional services, and so on), then computed each sector’s share of county employment for every county-year.

Two data issues limited the usable time window. First, even after Eckert et al.’s NAICS 2012 concordance, the 1990-1997 numbers still has a structural break because those years originate in SIC. National manufacturing shares came out around 28-33% before 1998 and then dropped to 16% in 1998, which is a definitional change. Second, the Census changed CBP reporting starting in 2017 by perturbing small cell values to protect confidentiality, and the readme specifically warns that 2017 and 2018 are ”fundamentally different from earlier periods.” I therefore restricted the merged covariate to 1998-2016, which gives 19 clean years of comparable data. That window still covers most of the structural transformation we want to document.

Table 6 shows population-weighted national employment shares for six selected sectors at five anchor years. Manufacturing fell from 15.8% in 1998 to 9.2% in 2016, a drop of 6.6 percentage points concentrated between 2000 and 2010 (the years commonly associated with the China trade shock). Healthcare rose from 12.9% to 15.7% and overtook manufacturing as the largest sector around 2007-2008. Retail stayed near 13% throughout. Professional services and education grew steadily. Agriculture stayed marginal in employment terms, which does not mean rural counties escaped the story, only that agricultural employment itself is small relative to the sectors that do the hiring.

Sector	1998 (%)	2003 (%)	2007 (%)	2012 (%)	2016 (%)
Manufacturing	15.8	12.5	11.0	9.6	9.2
Healthcare	12.9	13.7	13.9	15.8	15.7
Retail	13.3	13.2	13.0	12.9	12.7
Professional services	5.6	6.4	6.7	7.0	7.0
Education	2.2	2.5	2.7	3.0	3.0
Agriculture	0.2	0.2	0.2	0.2	0.2

Table 6: Population-weighted national employment shares by NAICS 2-digit sector. Manufacturing fell 6.6 pp while healthcare rose 2.8 pp and overtook it as the largest sector around 2007-2008.

Figure 2 plots the full annual series for these six sectors, so the crossover between manufacturing and healthcare is visible year by year. The pattern is important for the attainment story because the sectors that grew (healthcare, professional services, education) draw much more heavily on BA-and-higher workers than the sectors that shrank (manufacturing, agriculture). Counties whose 1998 employment was concentrated in manufacturing had a structural headwind against BA share growth over the next two decades. Counties concentrated in healthcare, education, and professional services had a structural tailwind. When we get to modeling the metro-rural attainment gap, the county-level industry mix is a candidate mechanism worth testing directly.

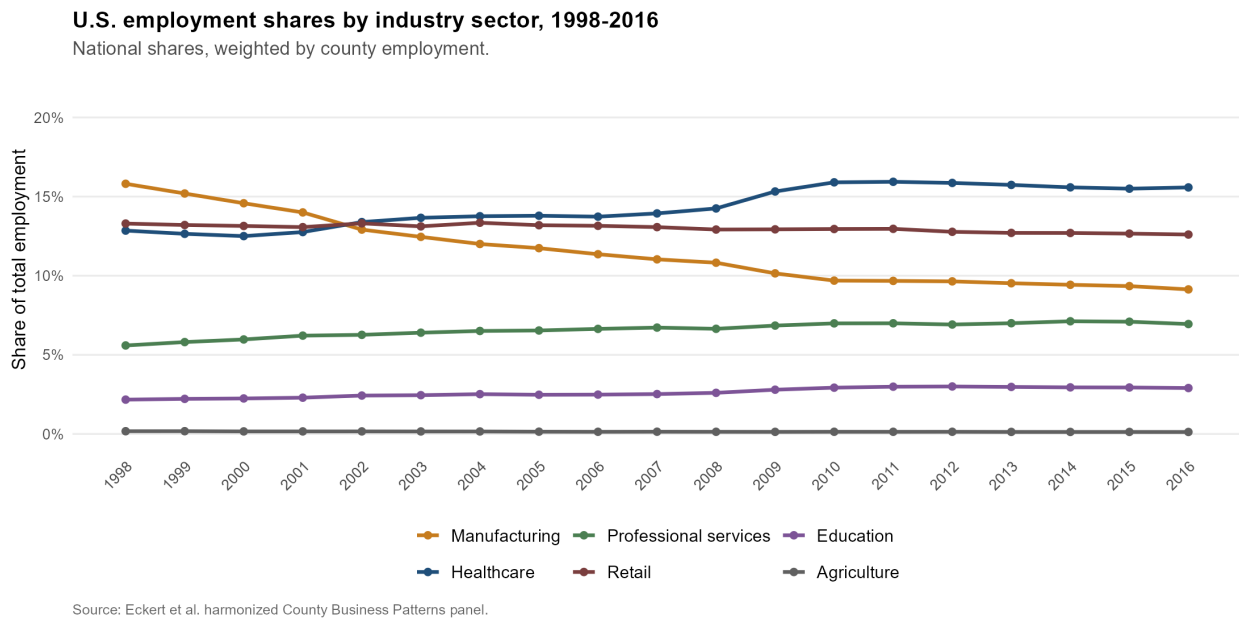


Figure 2: U.S. national employment shares by NAICS 2-digit sector, 1998-2016. Manufacturing fell steadily; healthcare rose and overtook it around 2007-2008.

File location

The CBP-merged panel is saved as `data/cleaned/us_ed_county_panel_long_1940_2024_full_cbp.rds` and `.dta`. The standalone sector-share panel is `data/cleaned/cbp_sector_shares_1990_2018.rds`. The build script is `code/22_parse_merge_cbp.R` and the figure script is `code/23_plot_sector_shares.R`.

Data, Code, and Figures Availability

Everything used in this report, including the raw and cleaned data files, codes, and figures are available in the [GitHub repository](#). Anyone with access can rerun everything from the raw files through the final outputs and reproduce these results.